TASK SPECIFICATION: P2-03 PHASE P2 — IDENTITY, SECURITY & CAMPUS CORE

Authentication API (Register, Login, Refresh, Logout) & RBAC

Git Branch: `feature/P2-03-authentication` · Target Pull Request: PR (feature/P2-03-authentication)

Core Objective & Technical Scope:
Stateless JWT authentication engine with 15-minute access tokens, rotating refresh tokens, role-based authorization guards, and endpoint rate limiting.

Priority Level	Critical	Complexity Rating	L
Execution Status	Done ✅	Target PR / Commit	`PR (feature/P2-03-authentication)`
Upstream Dependencies	P2-02		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Configure Spring Security with stateless session management and JWT authentication filter.
2. Implement JwtService supporting HMAC-SHA256 signature verification and expiration handling.
3. Implement AuthService handling registration, credential validation, token rotation, and revocation.
4. Create AuthController exposing /api/v1/auth/ endpoints with OpenAPI documentation.
5. Author security test suite verifying authentication matrix and token tampering prevention.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `common/security/*`
- `identity/api/AuthController.java`
- `identity/application/AuthService.java`
- `DTOs`
- `tests`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Registration creates user and registers initial vehicle atomically.
- Invalid credentials return 401 Unauthorized without user enumeration.
- Expired or tampered JWTs rejected immediately; refresh rotation revokes previous tokens.

VERIFICATION & TEST COVERAGE STRATEGY

- AuthServiceTest
- AuthControllerIntegrationTest
- SecurityMatrixTest
- TokenTamperTest